Pergamon

Spill Science & Technology Bulletin, Vol. 6, No. 2, pp. 145–152, 2000
Published by Elsevier Science Ltd.
Printed in Great Britain
1353-2561/00 $ - see front matter

PII: S1353-2561(00)00041-4

RESEARCH

Analysis of Methods Used in Spill Response Planning: Trajectory Analysis Planner TAP II

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Long records of geophysical forcing have been used in numerous studies to estimate a statistical distribution of oil spill scenarios. The resulting set of spill scenarios is then used as a basis for planning a robust response capability that should be able to handle all likely real spills. For model developers to be able to support these expectations there are a number of criteria that must be satisfied: (1) Models must develop and retain the data necessary to answer key response questions; (2) developers must understand the limitations in resolution imposed by the specific algorithms they use; and (3) the cardinality of the long geophysical records (with respect to modeled spill behavior) should be determined and the final collection of spill scenarios must span this set. This paper considers these specific constraints and discusses methods that can be used to quantify some aspects of the uncertainty in the output. Published by Elsevier Science Ltd.

Keywords: **Oil spill response, oil spill planning, oil spill trajectories, statistics**

Introduction

Planning a response to a possible oil spill is a complex business and typically brings together special interests from a wide variety of disciplines including engineers, resource trusties, ornithologists, property owners, government managers, representatives of the marine community, etc. In order for these groups to begin to form a unified plan, they must have a general understanding of the oil spill threat for which they are required to plan. Often, many of the members of the planning team have little or no background in physical oceanography and trajectory analysis procedures. The task for modelers who wish to support these planning groups is to present realistic ensemble information on potential spills and their consequences with respect to the various special interests represented by all the diverse members of the planning team. A more detailed discussion of this problem is given in Galt & Payton (1988).

The hazardous materials response division (HAZMAT) of the national oceanic and atmospheric administration (NOAA) has developed a user-friendly tool to support spill response planning (trajectory analysis planner (TAP)). It is an application and viewing engine for an area specific database that allows end users to view the probability of oil moving from a specified spill site to a designated target location. This project was carried out as a pilot study for two locations (Delaware Bay and San Francisco Bay), and the initial version was completed in the second quarter of 1998. After completing the pilot we worked with planners to evaluate their reception of the product and observe how they interpreted the statistics that they were given. In addition, we began a careful analysis of the statistical methods used in TAP so that we could provide the user with an estimate of the uncertainty of the information we were presenting and

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acknowledge potential gaps in scenario space that might not be covered in the initial geophysical data used to generate the scenario set.

The results of the usability testing lead us to the realization that the configuration of TAP did not provide some of the statistical analysis that would be the most useful to planners, and that the information that it did provide was often misinterpreted. In addition, users often overestimated the resolving power of the statistical methods, trying to address response problems in excessive detail, subsequently bogging down the entire planning process. The results of these studies has led to the reformulation of the TAP analysis procedure. The new version is referred to as TAP II and provides a greatly expanded and more powerful spill trajectory database that specifically addresses major questions of concern to spill response planners.

The rest of this work describes the statistical data base that is developed for TAP II and addresses the sensitivity issues that are important in presenting and using the data for response planning. The next section looks at the new data structure of the TAP II scenario set and explains how it is analyzed to address specific questions that are critical to spill response planning. Also investigated is the statistical relevance of representing individual spell scenarios with a finite number of representative oil particles (Lagrangian Elements). A simple statistical model is introduced to obtain quantitative estimates of expected errors associated with this facet of the model. We also investigate methods that are used to estimate the cardinality of long period geophysical forcing and trajectory modeling sets so that it is possible to estimate how many scenarios are required to span the situation space that planners need to consider.

Data Structure Development for TAP II

The original version of TAP (Galt & Payton, 1988) looked at thousands of trajectories linking spill sites and receptor sites, to develop the statistics of a spill from any particular location impacting any one of a number of high value target sites. The design called for modeling many fragments of many spills, resulting in a fast and efficient algorithm. The resulting database described the time-dependent build up of the probability that any particular site might be oiled and the average (over all spills) amount of oil that would be expected at that site. When subjecting the model to usability testing and watching its sensitivity to variations in local geometry, several problems became apparent.

The first problem was computational and related to the way receptor mode trajectories were processed. This is an inverse method and is sensitive to the grid resolution that is used at both the hit site (high value resource) and the launch site (spill location). For areas where a large-scale feature (such as San Francisco Bay) joins a smaller feature (such as Carquinez Strait), it becomes very difficult to maintain computational reliability without reducing the integration grids to the point where the efficiency of the method is negated.

The second problem was related to the averaging process used to count spill hits and the fact that we did not calculate correlation information about simultaneous hits that might occur on different target sites from the same spill. Although TAP did give information about how likely it was that a particular site might be oiled, it was unable to define over what fraction of the spills those hits might occur. For example, if a result indicated that 10% of the oil impacted a particular site, the user could not distinguish between one tenth of the spills bringing one percent of the oil, or one hundredth of the spills bring 10% of the oil. Most of the users tested misunderstood this subtlety and assumed that the percents given could be applied to the fraction of any spill. High numbers was assumed to mean more oil from more spills. The real difficulty was that we were calculating a well-defined statistic that was not of particular interest to spill response planners. A more detailed evaluation of the problem led to the understanding that there are two fundamental questions that a statistical set of spill trajectories must answer and that TAP should be redesigned to meet those needs.

The first question has to do with what fraction of the spills from a particular site will bring more than a specified amount of oil to a high value receptor site. To collect these statistics, we must model "all" of an ensemble of spills and record the amount of oil that each spill brings to the site in question. From these data it is possible to design a site protection strategy giving the amount of equipment to protect against the median spill, 80% spill, etc.

The second question has to do with how likely it is that more than one site will be hit by any individual spill, and, more specifically, what fraction of the spills from a particular site will hit n or more sites for which the cost or consequences of a hit is known. To collect these statistics, we must once again model "all" of each spill and record what fraction of that spill goes to each receptor site. From these data it is possible to calculate the fraction of spills that would lead to higher than a given cost or consequence summed over all of the receptor sites. This is referred to as shortfall analysis by the response community and provides estimates of the amount of resources that must be

staged locally and the time demands that must be put on additional equipment available through mutual aid agreements.

One difficulty with the computational scheme used in TAP was related to the specification of high value target sites. Originally, we asked the local planning committee for a list of sensitive sites. We found that this list was unstable as new combinations of planners worked on the problem, new sites were continually added to the list. In addition, there have been sizable GIS efforts gathering compilations of high value resources and shoreline types ranked against potential oil damage, for many planning areas. We felt that TAP II could represent a more complete analysis by using a segmented partition of the entire coast, rather than a pre-determined number of target sites. The new formulation in San Francisco provides hit statistics for about 180 contiguous 2 km segments. In this new format it is possible to analyze the shoreline segments against the regional environmental sensitivity index (ESI) data and include these as pre-compiled consequence or cost data lists. Associated with each of the shoreline segments will be a list of how much of each sensitive type (shoreline class, sea birds, endangered species, etc.) is present. For any of the spill scenarios we can immediately see, for example, the fraction of spills that would endanger more than a specific number of kilometers of class 10 (most sensitive) shoreline.

TAP II data structure

To summarize the data structure for TAP II, we can consider a large data cube (Fig. 1) that is calculated for each potential spill site within our region of interest. (In practice, cubes are calculated on an irregular grid and intermediate values are interpolated using a first-order triangular mesh.) A horizontal face in the cube represents the hit data from a single spill. Rows from left to right are the hits on each shoreline segments at a fixed time. Successive rows going back into the cube give the same type of hit data for later times from the same spill. A typical number of target shoreline segments is about 150. The time depth of the cube is between three and five days depending on the region to be modeled. The vertical faces in the cube represent a fixed amount of time from the start of the spill. On these time faces each column contains the hit data at a single shoreline segment for each of the set of spill scenarios. A row across the face of a time face gives the hit data at each shoreline segment for one particular spill scenario. This is the critical correlation data that are required for shortfall analysis.

The use of this cube data to answer the fundamental response questions that were outlined above can now be easily demonstrated. Given a spill (site, amount and type of oil) and a time horizon, the data of interest will be within a single time plane from one of the cubes in the database (Fig. 2). Using appropriate weathering information, the site hit data can be transformed into a specific amount of oil. For any site that might be identified, the oil arriving from each individual ensemble scenario will be in the column identified with that site. The column data are ordered from highest to lowest values and then plotted over a normalized abscissa (0–100%) with the ordinates being the amount of oil. From this presentation the planner can easily see what fraction of the modeled spill brings more than a particular amount of oil to the site at that time.

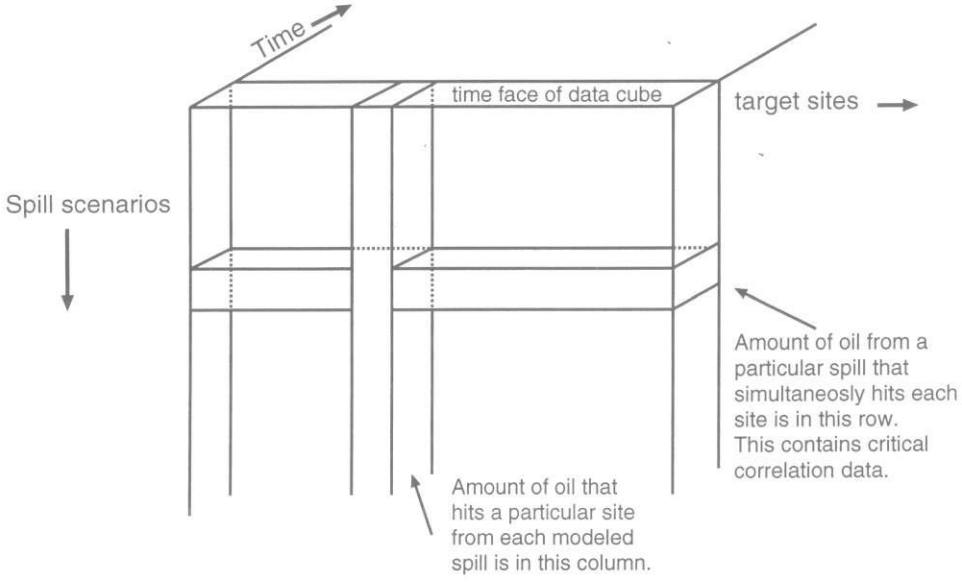


Fig. 1 Layout of the data cube giving information and correlation data for a number of scenarios all originating from the same source location.

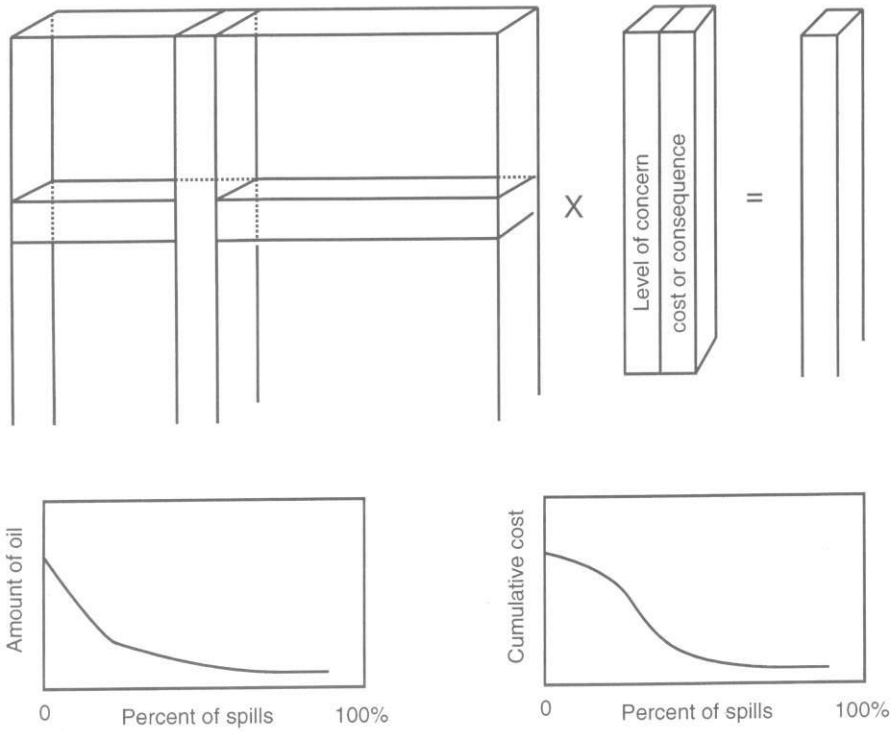


Fig. 2 Computational scheme on a time face of a data cube to give the site oiling analysis, and the cumulative cost or consequence estimates analysis.

To use the data for shortfall analysis, the planner would start with the same time plan from a cube and consider how each spill (row) distributes oil simultaneously over all the sites. Additional data would include a set of column vectors over the sites that list cost or consequence and a level of concern that specifies at what level costs are incurred. Using the level of concern as a Heaviside function (If the value in the time plane is greater than or equal to the LOC, it is multiplied by one; if the value is less than the LOC, it is multiplied by zero. Only those sites that have greater than the LOC of oil will contribute to the total.), the row in the time plane multiplies the column of cost or consequence and will give a new vector whose elements sum the total cost or consequence associated with each of the spills in the database ensemble. Once again, these costs can be ordered from highest to lowest and plotted over a normalized abscissa (0–100%) with the ordinates being cost or consequence. The planner can easily see what fraction of the spills will exceed a particular value of cost or consequence.

Uncertainty in TAP II

Each spill scenario is a trajectory computed by NOAA's on scene spill model (OSSM). OSSM is a Lagrangian element (LE) model, representing an oil spill as a number of individual particles that each represent a small fraction of the oil spilled. OSSM uses data about the local meteorological and oceanographic conditions to calculate the moving and spreading of the oil. Computational limitations require that a finite number of spill scenarios are modeled, and that a finite number of LEs are used in each simulation. The following is an assessment of the uncertainty introduced by these limitations.

Uncertainty from finite number of LEs

To determine the number of LEs required, the confidence intervals can be computed for the number of LEs that might impact a particular site at a particular time, for a particular run. The count of the number of LEs that hit a given site at a given time is an estimate of the probability of any individual LE from the spill hitting that site. A given LE can either hit the site of concern, or not. Thus the results are expected to come from the binomial distribution

$$P(X;p,n) = \binom{n}{X} p^X (1-p)^{n-X}, \quad (1)$$

where $P(X;p,n)$ is the probability of getting X hits, p the probability of a single LE hitting, and n is the total number of LEs. When a single spill is simulated the result is a single sample from this distribution, and we

are trying to determine p. A single run gives an estimate for p, $\hat{p}$, where

$$\hat{p} = \frac{X}{n}. \tag{2}$$

The question is: how accurate is this estimate?

If this experiment were run many times, the result would be a set of $\hat{p}$ that would come from the binomial distribution (Eq. 1). Confidence intervals for $\hat{p}$ can be computed using the F distribution (Zar, 1984). The lower confidence limit is

$$L_1 = \frac{X}{X + (n - X + 1)F_{\alpha(2),\nu_1,\nu_2}}, \tag{3}$$

where

$$\nu_1 = 2(n - X + 1), \tag{4}$$

and

$$\nu_2 = 2X. \tag{5}$$

The upper confidence limit is

$$L_2 = \frac{(X + 1)F_{\alpha(2),\nu_1',\nu_2'}}{n - X + (X + 1)F_{\alpha(2),\nu_1',\nu_2'}}, \tag{6}$$

where

$$\nu_1' = 2(X + 1) = \nu_2 + 2, \tag{7}$$

and

$$\nu_2' = 2(n - X) = \nu_1 - 1. \tag{8}$$

Using these relationships, it can be said that there is a $(1 - \alpha)\%$ chance that for a given $\hat{p}$, p is between L_1 and L_2. As an example, if 1000 LEs were used ($N = 1000$), and five hit a given site at a given time ($X = 5$), the 95% confidence limits ($\alpha(2) = 0.05$), are

$$L_1 = 0.0016,$$
$$L_2 = 0.0116.$$

While 0.5% of the LEs were counted hitting that site, there is a 95% chance that the real value is between 0.16% and 1.16%. To put it in understandable units, if 1000 barrels of oil were spilled, we can say that we are 95% sure that between 1.6 and 11.6 barrels of oil would have hit that site with that particular set of meteorological and oceanographic conditions.

Table 1 shows the results of computing the 90% confidence limits for 1% and 5% for a variety of different numbers of LEs. Using 100 LEs results in confidence limits that do not distinguish between 0.05% and 4.66%. As 1% of an oil spill is likely to be a significant amount of oil, this is clearly not accurate enough. It was determined that the confidence limits provided by 1000 LEs were reasonable, and the computational effort increases linearly with the number of LEs so the added accuracy obtained by using more could not be justified.

Table 1 Confidence limits for amount of oil recorded at a site, computed at 90% confidence

N	Lower limit	% of spill	Upper limit
100	1.99	5	10.23
500	3.50	5	6.91
1000	3.92	5	6.28
1500	4.11	5	6.03
2000	4.22	5	5.88
100	0.05	1	4.66
500	0.40	1	2.09
1000	0.54	1	1.69
1500	0.62	1	1.54
2000	0.66	1	1.45

Uncertainty from finite number of modeled spills

A far more difficult problem is assessing the uncertainty associated with the finite number of spills modeled. A simple approach is to examine the distribution of the amount of oil that impacts each given site. If this distribution can be parameterized, standard statistical techniques could be used to determine confidence limits on the parameters. While this would provide an assessment of the accuracy of the estimates at each site, it would provide no information as to how accurately the correlations between sites have been determined.

The correlations between sites are critical for shortfall analysis. Assessment of the uncertainty of the correlations between sites requires that the entire data set be considered together. Each spill simulation results in a single point in a N-dimensional space, where N is the number of receptor sites. Each receptor site is an axis of the space, with the fraction of the oil spilled hitting that site the value along that axis. A separate distribution of points in this space is generated for each time considered. The goal is to determine how accurately a set of modeled spills defines the distribution of points within this space.

As an example, the shoreline of San Francisco Bay was divided into 180 receptor site segments, each encompassing about 2 km of shoreline. This results in a huge space, so that fitting the data to a parameterized distribution is essentially impossible. The problem is simplified somewhat by the fact that there may be some sites that are never impacted by a spill from a given source, but after a few days, the oil can spread very far. In San Francisco Bay, five days after a spill in the Golden Gate, 31 sites had never been hit by any of 500 simulations, reducing the dimensionality of the space to 121, still an unmanageable number.

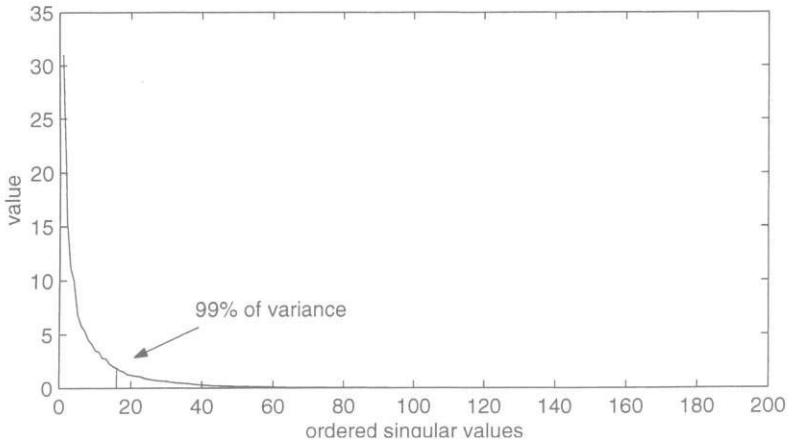


Fig. 4 Plot of singular values of SVD of simulations started at the Golden Gate.

of the separate subspaces, resulting in two slightly different representations of the set of spills

$$\mathbf{X}'_1 = \mathbf{X}_3 \cdot \mathbf{V}'_1 \cdot \mathbf{V}'^T_1,$$
$$\mathbf{X}'_2 = \mathbf{X}_3 \cdot \mathbf{V}'_2 \cdot \mathbf{V}'^T_2. \quad (11)$$

Each row of these matrices is an individual spill projected into each of the two subspaces. Comparing a row in $\mathbf{X}'_1$ to the same row in $\mathbf{X}'_2$ results in an assessment of how similar the two subspaces, $\mathbf{V}'_1$, and $\mathbf{V}'_2$ are. If all the spills in one subscpace correlate well with all the spills in the other subspace, then the subspaces are similar.

Table 2 contains the results of performing this analysis with samples of various sizes. The correlation coefficient (r) between each spill in one subspace and the same spill in the other subspace has been computed, and the minimum, mean, and maximum correlation coefficients are presented. With as few as 50 spills, the correlations are quite high. For this application, however, it is important that the correlations are particularly strong. In most instances, the correlation coefficient is used to assess whether two variables have a relationship with one another. In this case, there is no question that a spill projected into one space is related to the same spill projected in another space. The question is whether the detail of the distribution of spills is the same. Establishing a standard of a mean r of at least 0.99, and that the minimum r of 0.95 requires that 500 spills be modeled for that location.

Table 2 Correlations between the same spills projected into two different subspaces

Number of spills	Min. r	Mean r	Max. r
50	0.89284	0.97812	0.99662
100	0.93675	0.98886	0.99830
200	0.93999	0.99388	0.99882
300	0.93060	0.99418	0.99961
400	0.94097	0.99565	0.99980
500	0.95842	0.99567	0.99964

In the case of San Francisco Bay, spills started at the Golden Gate are subject to strong tides and complex geometry, and thus resulted in a larger variety of scenarios than in other locations. In most other regions, fewer than 500 spills resulted in excellent correlations, so using 500 spills everywhere would result in better than the required accuracy in most locations.

While the previous analysis indicates that the representative subsamples span the same space, a stronger test of similarity would be to determine whether the samples have the same distribution within each of the principle components. We are currently working on developing methods to examine this more stringent requirement.

Conclusion

The HAZMAT of the NOAA has developed a second generation user friendly tool to aid in oil spill response planning (TAP II). The second version of this tool was developed to address limitations in the first version (TAP) that were discovered during observations of users. TAP II utilizes a new data structure for a statistical database of simulated spills that preserves correlation information between different locations that might be affected by an oil spill. This new data structure provides planners with the information required to perform shortfall analysis of response equipment and personnel. Analysis of the computed data set provided by many simulations provides estimates for some of the uncertainty in the resulting statistics.

References

Galt, J.A., Payton, D.L., 1988. Development of quantitative methods for spill response planning: A trajectory analysis planner. Science and Technology Bulletin 5 (1).

Preisendorfer, R.W., 1988. Principle Component Analysis in Meterology and Oceanography. Elsevier, Amsterdam.

Press, W.H., Flannery, B.P., Teukolsky, S.A 1986. Numerical Recipes. Cambridge Uni bridge.

Zar J.H, 1984. Biostatistical Analysis. Prenti Cliffs, NJ.